

AI Lie Detector: Distinguishing Human vs. AI-Generated Text

A MACHINE LEARNING APPROACH TO LINGUISTIC PATTERN ANALYSIS

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The Problem: Why AI Detection Matters

Academic Integrity

Misinformation

Legal/Forensic

Content Moderation

Project Objectives & Research Question

Dataset Curation

Feature Identification

Model Development

Testing it manually on command line interface

System Architecture Overview

Data Collection Module

Feature Engineering Core

Model Training Pipeline

Prediction Interface

Dataset Creation Strategy

Combine real essays with standard AI-detection datasets Generate additional synthetic texts to balance topics

Roughly 2,500 human and 2,500 AI samples after cleaning

Stratified 80/20 train–test split

Merge real and synthetic data to broaden style coverage

Feature Engineering: The Linguistic Fingerprints

Unique Word Ratio

AI Word Frequency

Sentence Length Variation

Average Word Length

Punctuation Density

Conjunction Ratio

Paragraph Structure

Readability Score

Model Selection & Training

Handles Non-linearity

Feature Importance

Robustness

Probability Outputs

Results & Performance Metrics

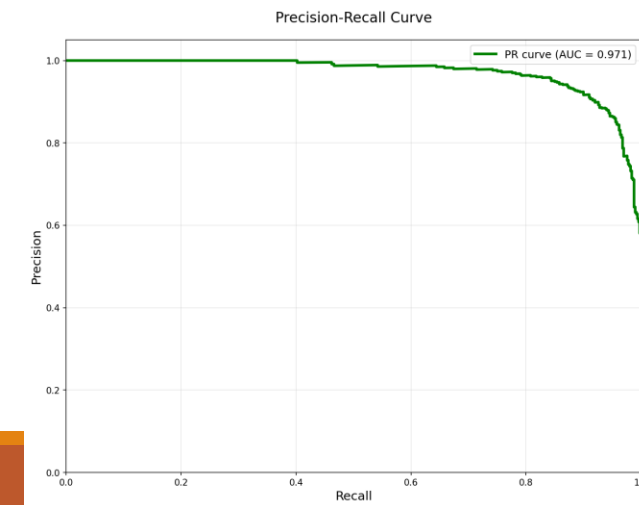
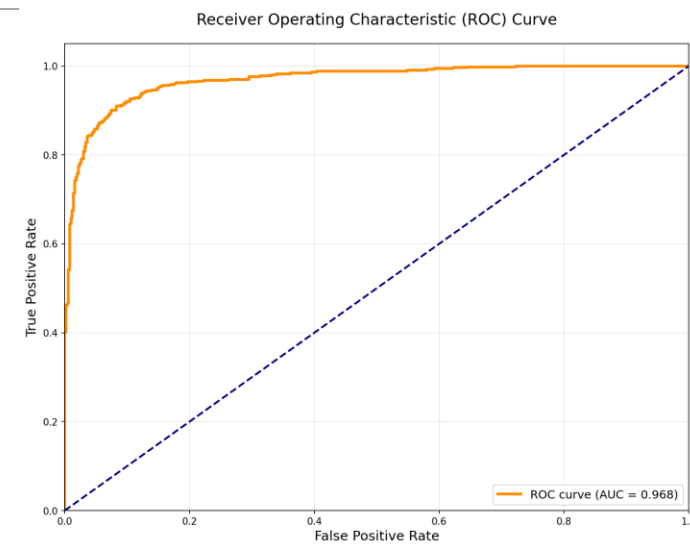
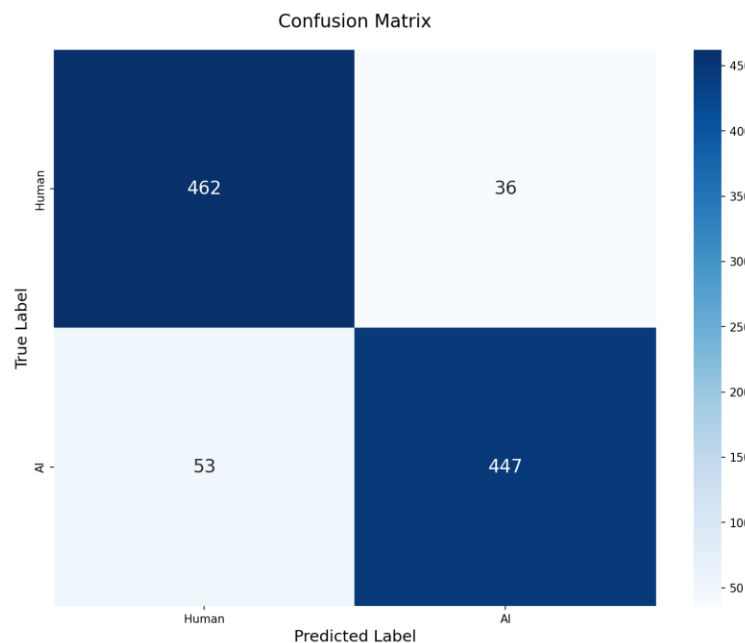
Test accuracy: 91.1% on 998 held-out samples

Precision: 92.6%, Recall: 89.4%, F1-score: 91.0%

ROC AUC: 0.968

False positive rate (human $\rightarrow$ AI): 7.2%

Average inference time: $\sim$ 25 ms per text



Feature Importance Analysis

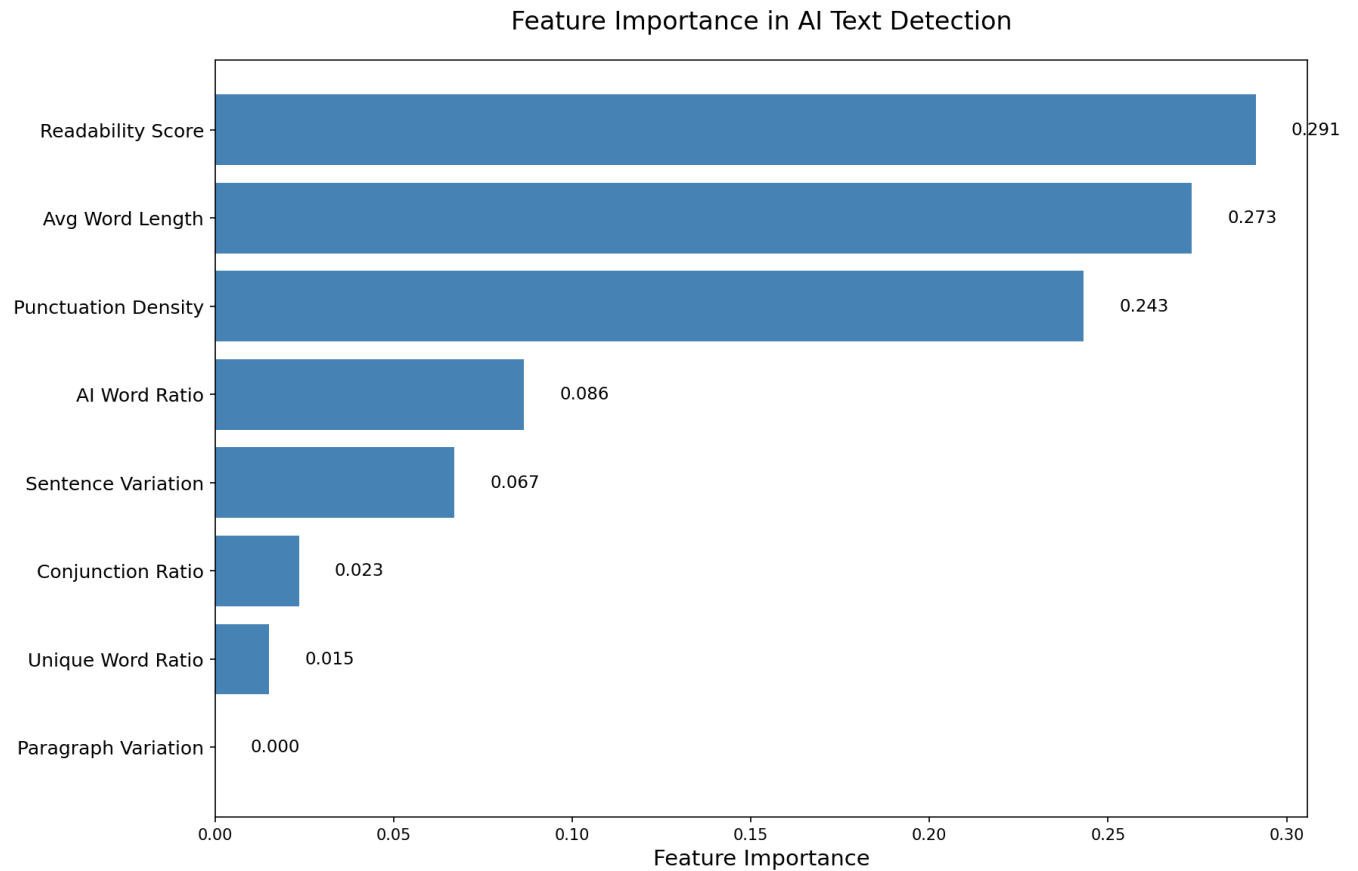
Readability Score (~29% importance)

Average Word Length (~27%)

Punctuation Density (~24%)

AI Word Ratio (~9%)

Sentence Length Variation (~7%)



Case Studies & Examples

Example 1: Clearly AI Text

Read the example text: "The multifaceted nature of contemporary computational paradigms necessitates a comprehensive reevaluation of traditional algorithmic frameworks..."

Analysis: AI written, confidence : 51.6%, AI Probability: 51.6%, Human Probability: 48.4%

Example 2: Clearly Human Text

Read the example text: "I can't believe how bad traffic was today. Seriously, took me an hour to go 5 miles!"

Analysis: Human-Written, Confidence: 85.1%, AI Probability: 14.9%, Human Probability: 85.1%

Example 3: Borderline Case

Read the example text: "Honestly, I think the main issue is communication. However, one must consider broader societal implications..."

Analysis: Prediction: Human-Written, Confidence: 50.9%, AI Probability: 49.1%, Human Probability: 50.9%

Conclusion

Effective System

Novel Insights

Transparent Approach

Accessible Tool

Final Reflection

Q & A
